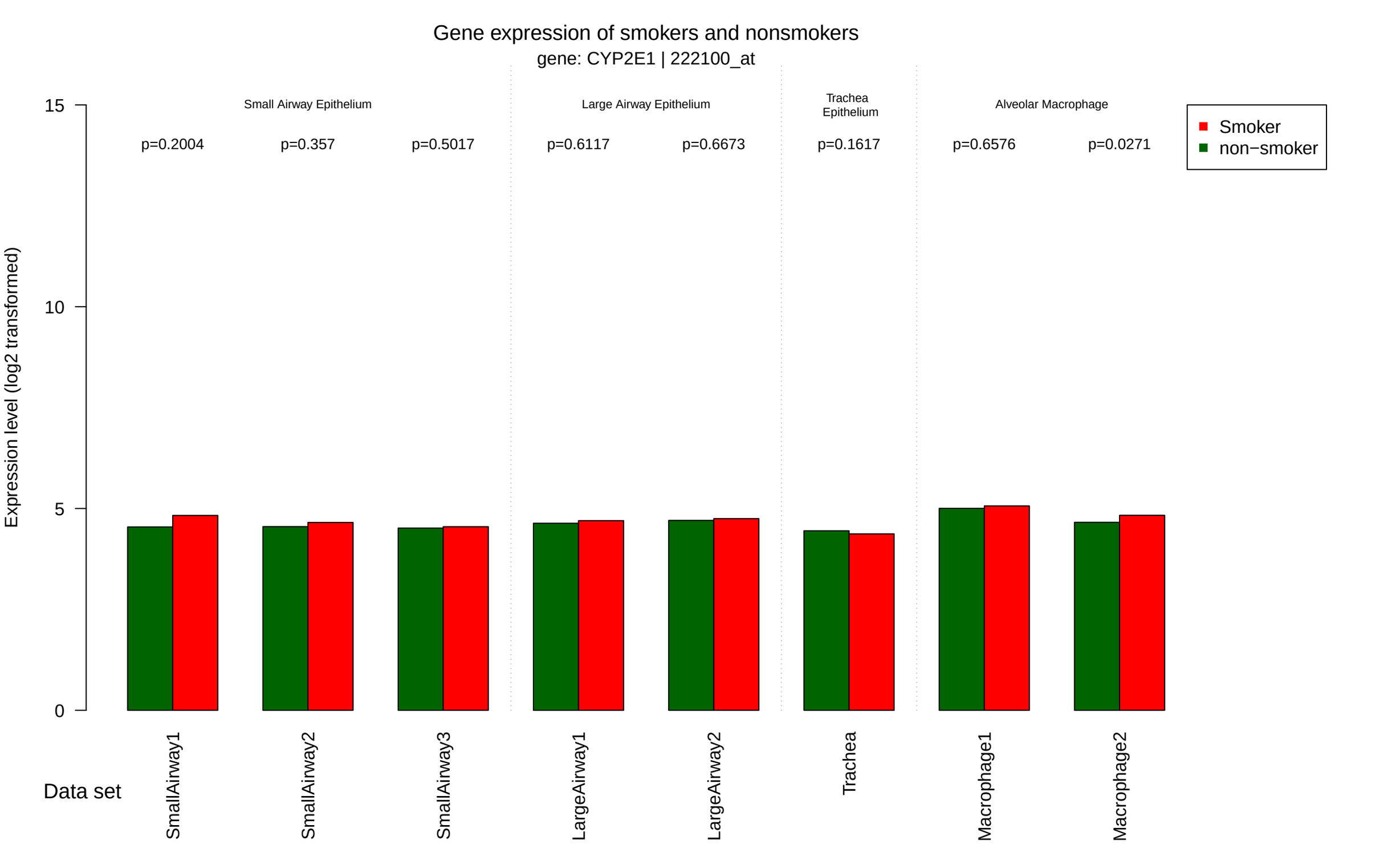
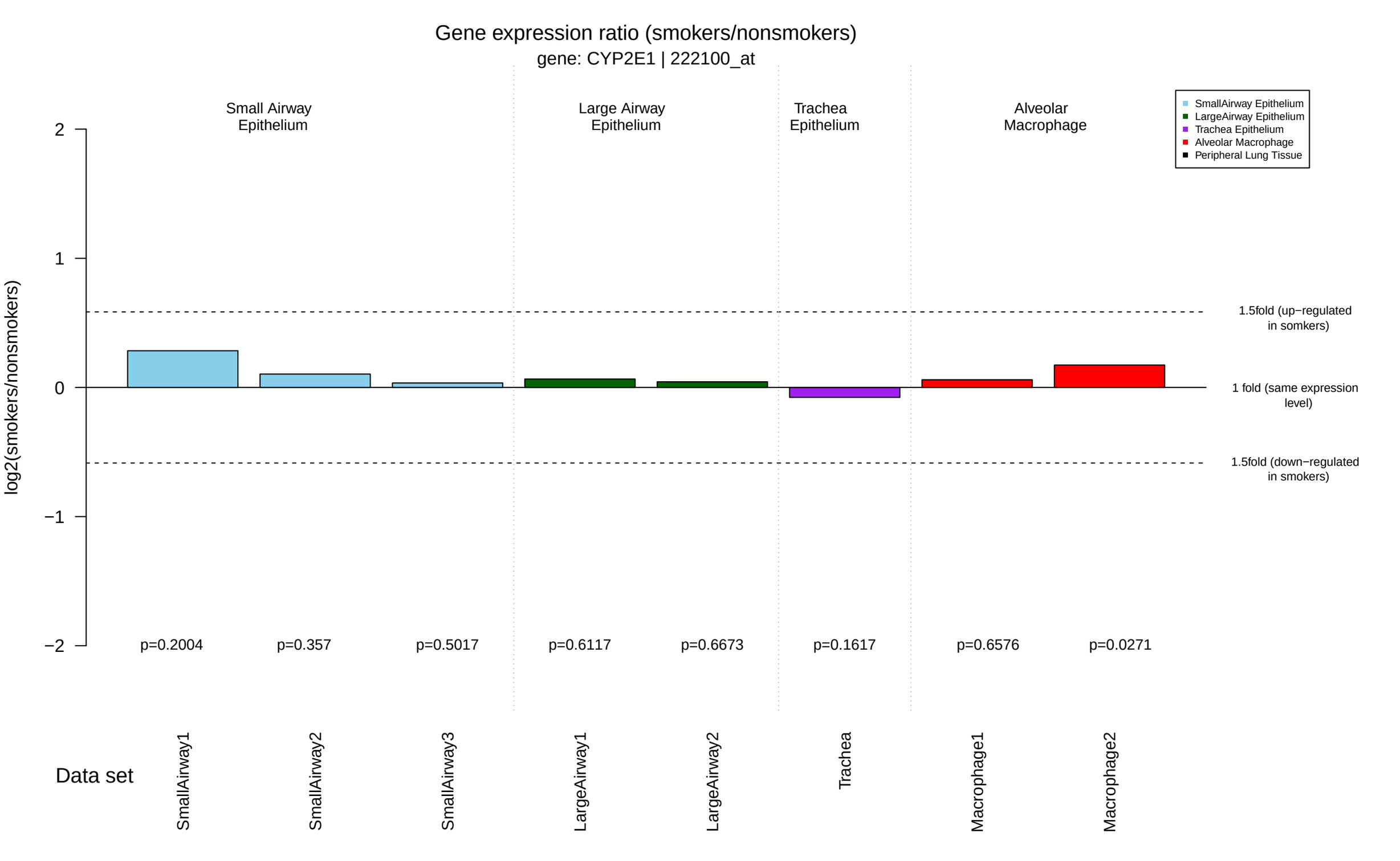
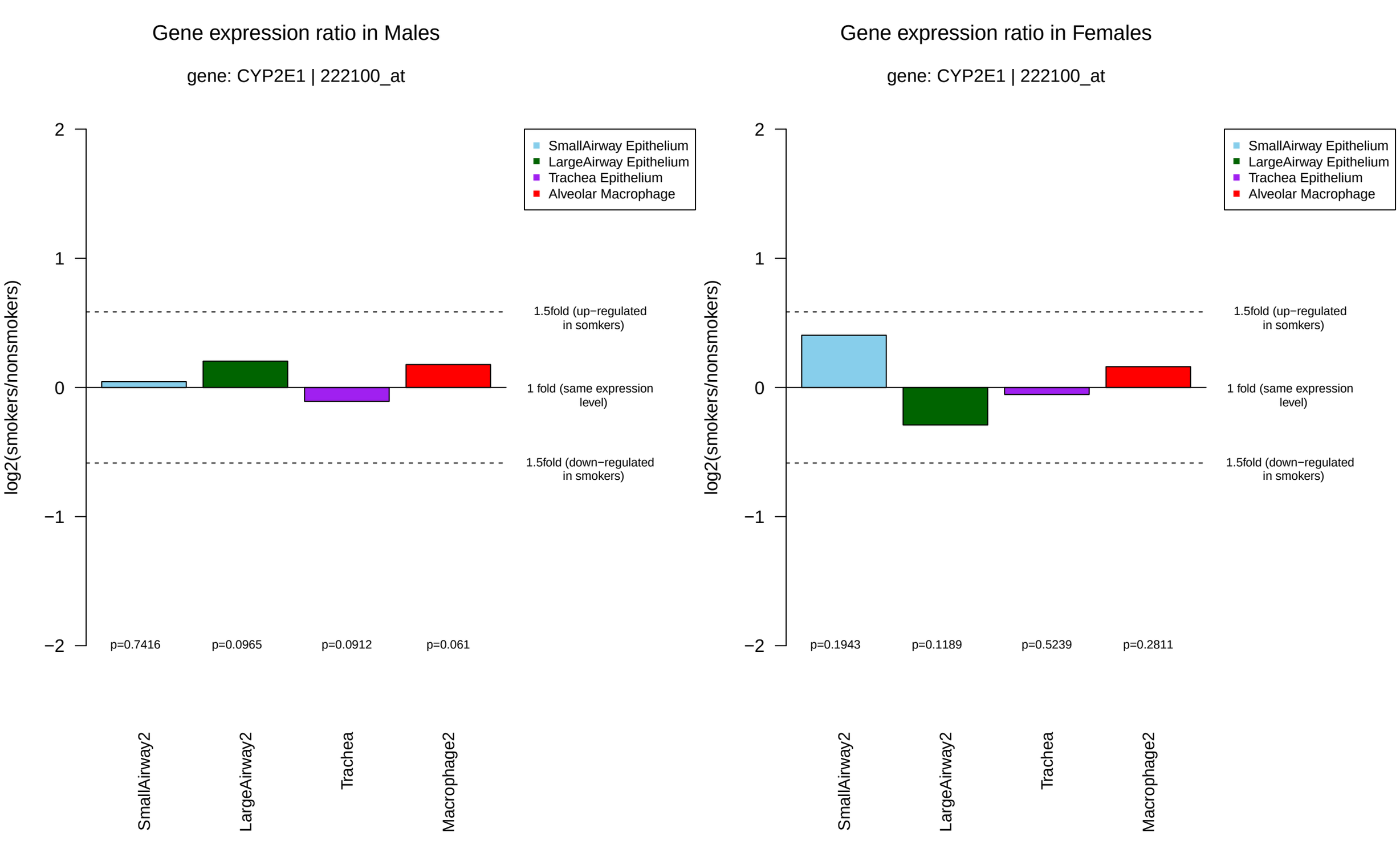
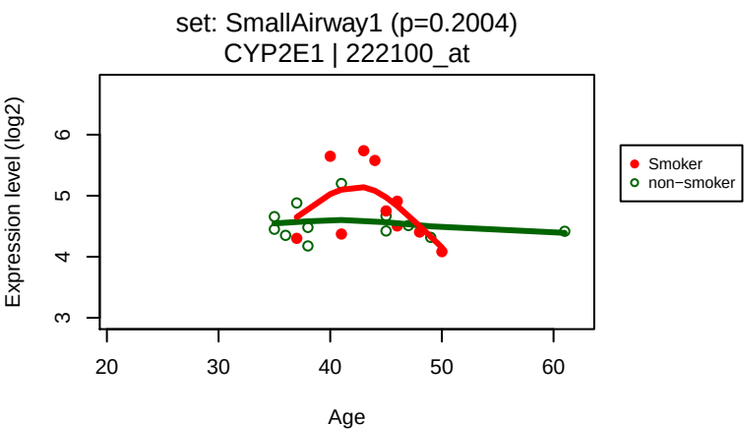
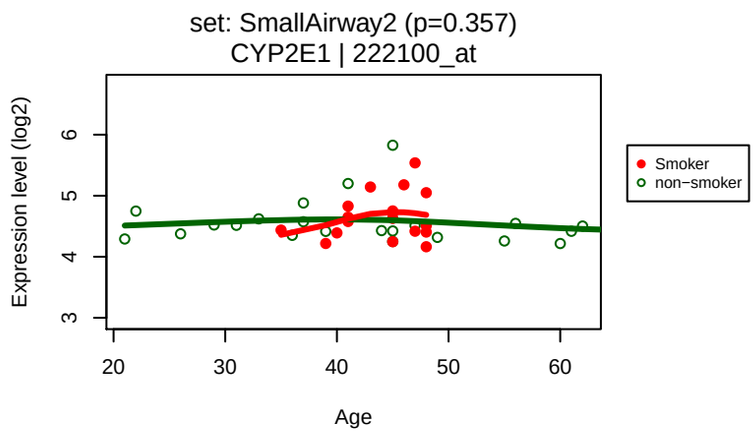
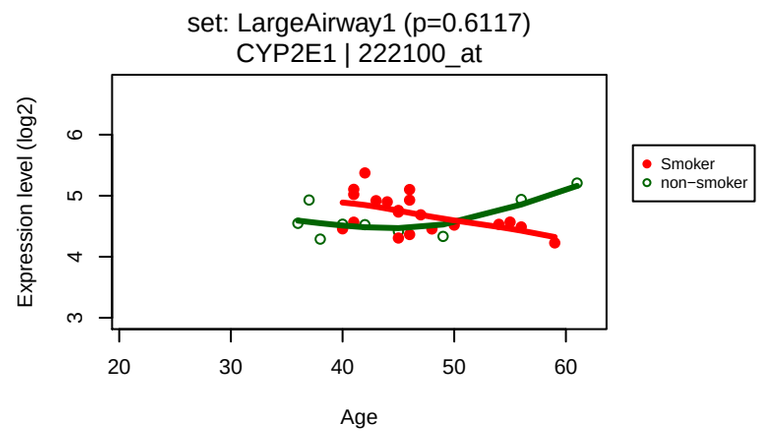

Gene expression of smokers and nonsmokers
gene: CYP2E1 | 222100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2004
p=0.357
p=0.5017
p=0.6117
p=0.6673
p=0.1617
p=0.6576
p=0.0271
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

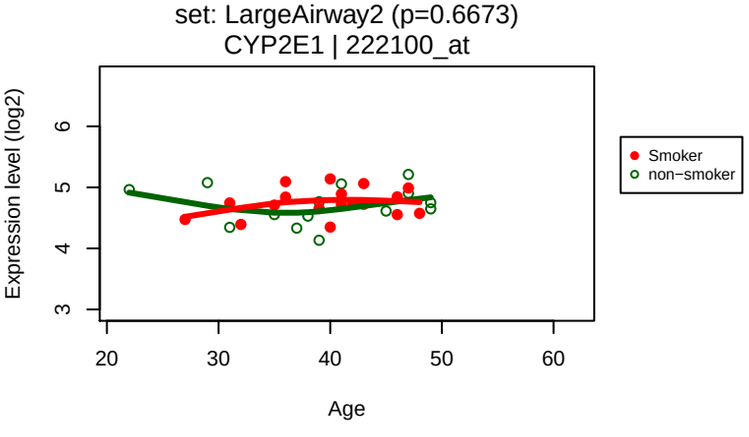

Gene expression ratio (smokers/nonsmokers)
gene: CYP2E1 | 222100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2004
p=0.357
p=0.5017
p=0.6117
p=0.6673
p=0.1617
p=0.6576
p=0.0271
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

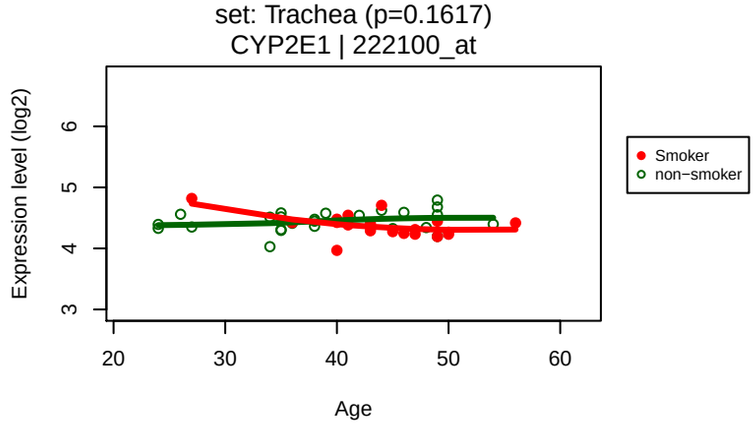

Gene expression ratio in Males
gene: CYP2E1 | 222100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7416
p=0.0965
p=0.0912
p=0.061
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2E1 | 222100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1943
p=0.1189
p=0.5239
p=0.2811
SmallAirway2
LargeAirway2
Trachea
Macrophage2

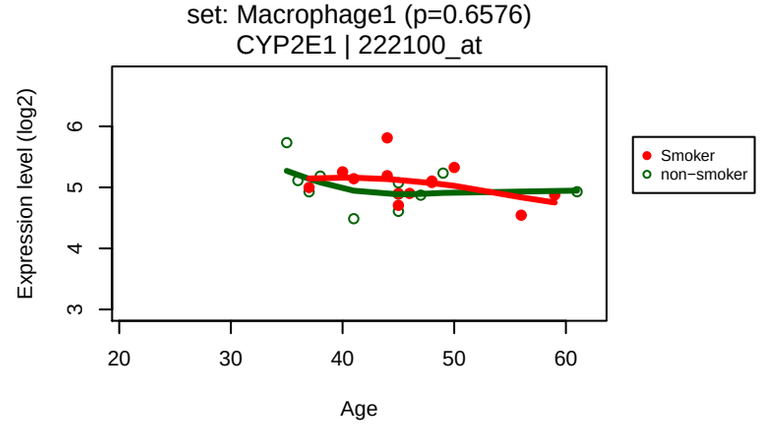

set: SmallAirway1 (p=0.2004)
CYP2E1 | 222100_at
Expression level (log2)
Age
Smoker
non−smoker

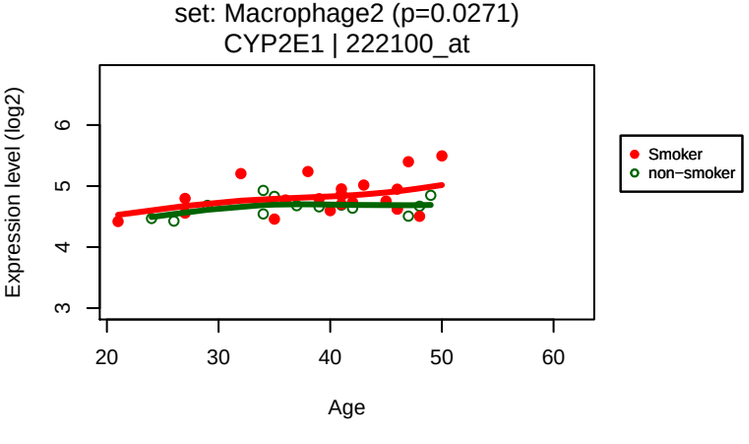

set: SmallAirway2 (p=0.357)
CYP2E1 | 222100_at
Expression level (log2)
Age
Smoker
non−smoker

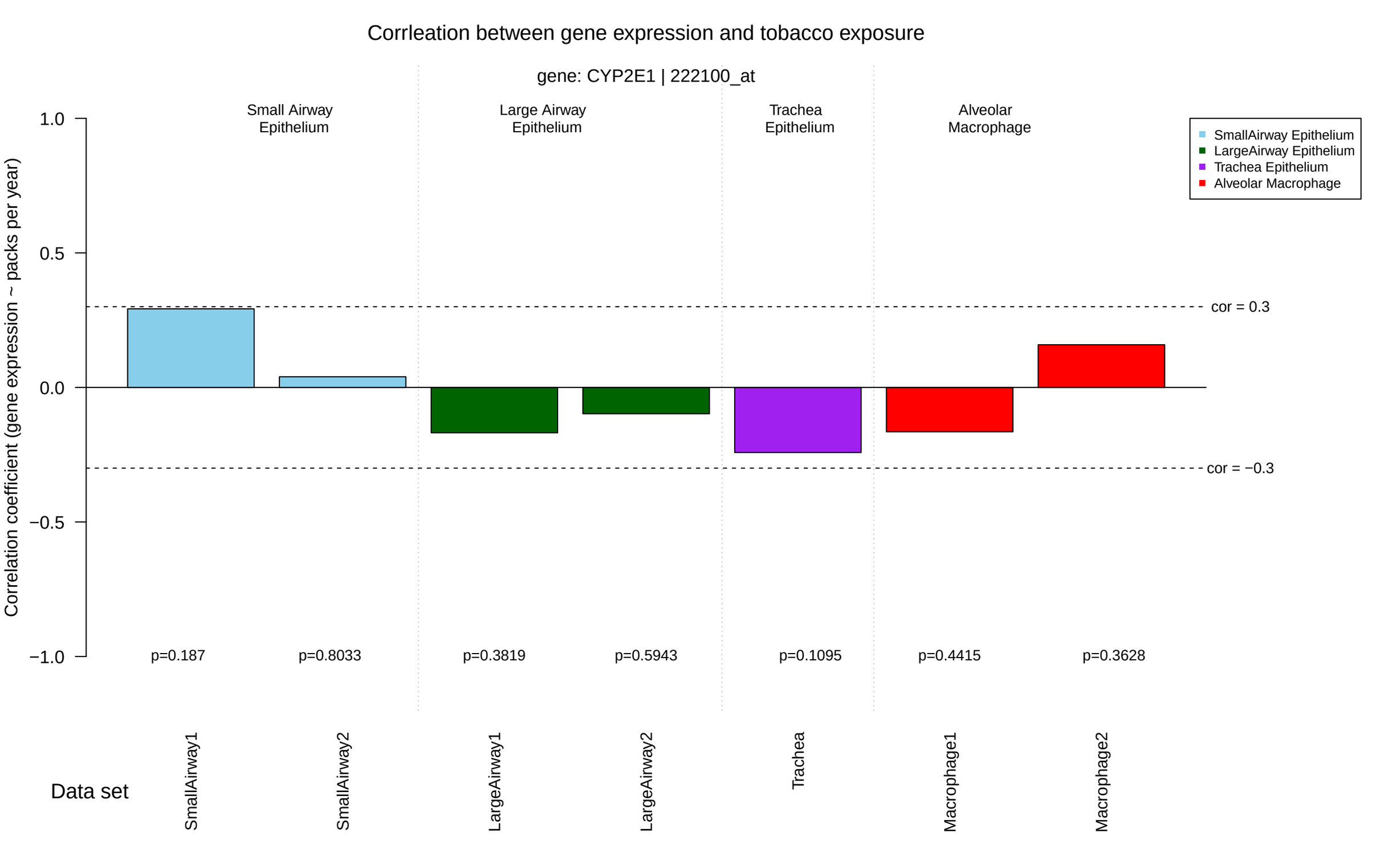

set: LargeAirway1 (p=0.6117)
CYP2E1 | 222100_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6673)
CYP2E1 | 222100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1617)
CYP2E1 | 222100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6576)
CYP2E1 | 222100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0271)
CYP2E1 | 222100_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2E1 | 222100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.187
p=0.8033
p=0.3819
p=0.5943
p=0.1095
p=0.4415
p=0.3628
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2